

Airport Air Quality Management

Protecting Local Air Quality and Community Health

Hun – Group 3

Environmental Aviation Course

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Why Airport Air Quality Management Is Important

Airports are major **local** sources of air pollution, mainly from activities during the Landing-Take-Off (LTO) cycle (below ~3,000 ft).

Key pollutants include:

- NO_x (nitrogen oxides) – causes smog and acid rain
- CO (carbon monoxide)
- HC/VOCs (hydrocarbons / volatile organic compounds)
- nvPM (non-volatile particulate matter, including ultrafine particles)

These can lead to respiratory problems (e.g., asthma), cardiovascular issues, and poorer air quality for communities near airports. The goal is to reduce emissions to meet standards while allowing responsible airport growth.

Main Emission Sources

- **Aircraft engines** – largest during taxi, take-off, landing (exhaust emissions)
- **Ground Support Equipment (GSE)** – diesel tugs, loaders, GPUs (high from idling)
- **Ground access traffic** – cars, taxis, buses (often biggest landside source)
- **Stationary sources** – fuel storage, generators, heating/cooling
- Other: construction dust, brake/tire wear, de-icing

Identifying sources enables targeted, cost-effective control measures.

ICAO Recommended Management Process

From ICAO *Airport Air Quality Manual* (Doc 9889) – a clear 5-step framework:

1. **Emissions Inventory** – quantify emissions by source
2. **Dispersion Modeling** – predict pollutant spread (weather, terrain)
3. **Impact Assessment** – compare to standards, identify hotspots
4. **Mitigation Planning** – select best actions (cost-benefit, stakeholders)
5. **Monitoring & Review** – measure real levels, verify improvements

This systematic approach ensures data-driven, compliant decisions.

Key Best Practices

- **Aircraft:** Use ground power & PCA (instead of APU); single-engine taxiing (saves 25–40% fuel/NOx); electric tugs (e.g., TaxiBot)
- **GSE:** Transition to electric/hybrid; reduce idling; optimize routes
- **Ground access:** EV charging stations; public transport incentives; solar parking

Integrate into airport master plan and Environmental Management System (EMS).

Real-World Examples

- Brisbane Airport: Electric buses saved 1.2 million liters diesel
- Delhi Airport: TaxiBot saved >52,000 liters fuel & 723 kg NOx

These demonstrate measurable air quality improvements and cost benefits.

Conclusion

Airport air quality management protects community health by reducing local pollutants. Using ICAO's proven process and practical steps like electrification and operational changes, airports can achieve regulatory compliance, support sustainable growth, and act as responsible neighbors. Proactive management benefits both the environment and airport efficiency.

Reference

International Civil Aviation Organization. (n.d.). *Air quality management at airports*.
<https://www.icao.int/sites/default/files/sp-files/environmental-protection/Documents/ecoairports/AIR%20QUALITY%20MANAGEMENT%20AT%20AIRPORTS.pdf>